ELEC70047
EE9-FPN1-02

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART IV: MEng and ACGI

HVDC TECHNOLOGY AND CONTROL

Tuesday, 13 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Question 1

a) State the factors that dictate the upper and lower limit of gate delay and gate advance angles at the rectifier and inverter of an LCC-HVDC link.

[3]

b) Explain the reasoning behind the allocation of control duty in a point-to-point LCC-HVDC link where the rectifier end is responsible for current regulation while the inverter end maintains the direct voltage under normal conditions.

[3]

c) This question relates to the control response of the rectifier and inverter of an LCC-HVDC link following a short circuit fault. Explain what initiates the control action and how the control responds when the fault occurs at the:

i) DC line/cable

[3]

ii) Inverter side AC system

[3]

iii) Rectifier side AC system

[4]

d) Explain why the inverter end of an LCC-HVDC link is vulnerable when connected to a weak AC system. Consider a Thevenin equivalent of the AC system at the point of connection with the inverter for your explanation.

[5]

e) Derive the expression for the ideal no-load direct voltage of a 6-pulse line commutated converter in terms of the RMS value of the line-to-line AC-side voltage. Consider the AC-side phase-to-neutral voltages $e_a = E_m \cos(\omega t + 60°)$ and $e_b = E_m \cos(\omega t - 60°)$ where E_m is the amplitude and ω is the grid frequency in $\mathrm{rad/s}$.

[4]

Question 2

Use all calculations on a per-pole basis for this question.

A point-to-point LCC-HVDC link interconnects two separate 400 kV AC systems. The HVDC link is rated at 2000 MW and $\pm$500 kV and uses a standard 12-pulse converter arrangement for each pole at both ends. The resistance of each pole of the HVDC line is 2.5 Ω. The AC-side voltage at both converter stations is maintained at their rated value. The turns ratio of the converter transformers is 0.5 with the low voltage windings on the converter side. Consider this turns ratio to be fixed throughout. The parameters of each 6-pulse converter used at both ends are given in Table 2.1 below:

Table 2.1: Parameters of each 6-pulse converter used at both ends of the LCC-HVDC link

Minimum gate delay angle	α_{min}	4°
Minimum extinction advance angle	γ_{min}	17°
Commutation resistance	R_c	5 Ω
Current margin	I_m	15%

a) In normal condition, the rectifier end of the HVDC link maintains the rated direct current while the inverter end operates with the minimum extinction advance angle. Under this base case, calculate:

i) Gate delay angle (in degrees) at the rectifier

[5]

ii) Reactive power consumed by the rectifier and the inverter

[4]

iii) Commutation overlap angle (in degrees) at the rectifier

[2]

b) If the AC-side voltage at the rectifier end drops to 92% of its rated value, calculate:

i) Extinction advance angle (in degrees) at the inverter

[4]

ii) Reactive power consumed by the rectifier and the inverter

[3]

iii) Commutation overlap angle (in degrees) at the inverter

[2]

c) Consider the LCC-HVDC link operates in the base case as in part 2(a). If the AC-side voltage at the inverter end drops to 94% of its rated value, calculate the gate delay angle (in degrees) and the reactive power consumed by the rectifier and the inverter.

[5]

Question 3

a) Answer the following questions related to the AC-DC converter shown in Figure 3.1. The converter parameters are summarised in Table 3.1.:

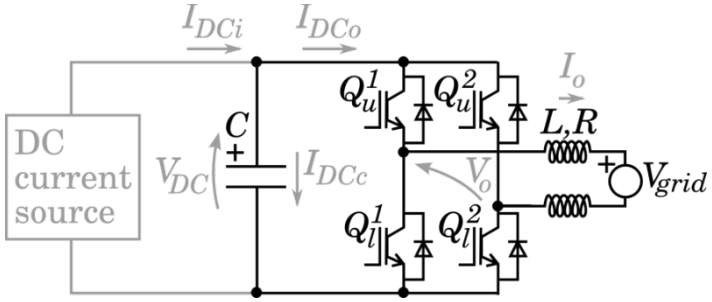


Figure 3.1 – Diagram of a single phase inverter connected to an AC grid

Parameter	Value	Units	Brief description
V_{grid}	230	V rms	Rated AC grid voltage
V_{DC}	400	V	Rated DC link voltage
P_{rated}	1.5	kW	Rated active power delivered to the grid
f_{rated}	50	Hz	Rated frequency of the AC grid
R	0.45	Ω	Equivalent resistance of bridge plus the filter
L	10	mH	Inductance of the filter

Table 3.1 – Diagram of a single-phase inverter connected to an AC grid

(i) Answer the following questions related to the following operating points:

- For an operating point with zero current at the output of the converter. What is the output voltage of the converter, V_o? What is the minimum V_{DC} voltage required to apply V_o?

[3]

- For rated output power ($P_{grid}=P_{rated}$, $Q_{grid}=0$). What is the output current? How much reactive power does the converter bridge have to feed to the output filter? What are the approximate conduction losses?

[3]

- For a given operating point where the 100 Hz component of I_{DCo} is 3.75 A of amplitude (ie half of peak-to-peak). What is the minimum capacitance, C, the DC bus would need for the 100 Hz ripple of V_{DC} caused by this current to be less than 4V of amplitude?

[4]

(ii) Regarding the current controller of the converter, we are told that the closed-loop transfer function between the current reference of the converter and the current is:

$$\frac{I_o}{I_o^*}(s) = \frac{1}{\tau s + 1}$$

with τ=3 ms.
Calculate the amplitude of the current and the phase delay compared to the current reference if we gave the controller a reference of 1 A at 50 Hz AC.

[3]

(iii) Write the Laplace transform of V_{DC} as a function of I_{DCo} and I_{DCi}.

[1]

b) Answer the following question related to the AC-DC converter shown in Figure 3.2. The converter parameters are summarised in Table 3.2

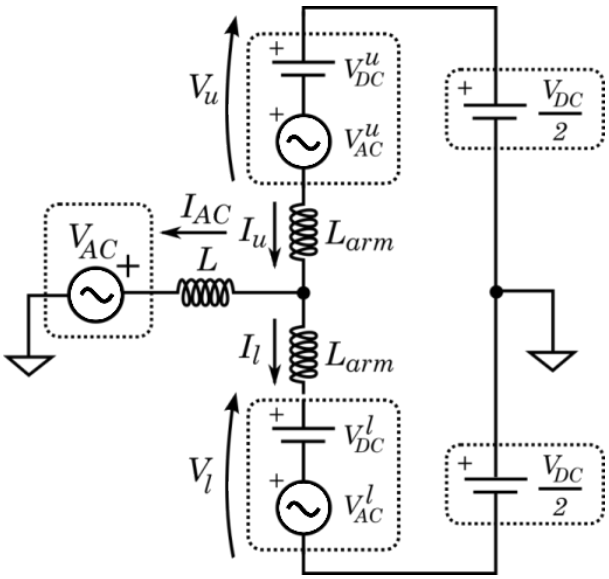


Figure 3.2 – Equivalent circuit of a two-arm single-phase AC/DC modular multilevel converter

Parameter	Value	Description
V_{DC}	120 kV	DC network voltage.
V_{AC}	38 kV	AC network voltage (RMS).
f_{AC}	50 Hz	Frequency of the AC network
S_{rated}	50 MVA	Rated power.
X_{arm}	17.8 mH (5.6 Ω at 50 Hz)	Arm reactor inductance.
X_L	4.58 mH (1.44 Ω at 50 Hz)	Phase reactor inductance
R_{arm}, R_L	0.28 Ω	Equivalent series resistances

Table 3.2 – Parameters of the two-arm single-phase AC/DC modular multilevel converter.

i) Calculate the AC and DC components of the currents of the converter (I_{AC}, I_u and I_l) and the voltages applied by the converter (V_u and V_l) when delivering rated power (P_{AC}=50 MW, Q_{AC}=0) under a stable operating point (ie no internal energy drift).

[6]

ii) We are told that the currents of the arms of the converter are: $\bar{I}_{AC}^u$=25 A, I_{DC}^u=15 A, $\bar{I}_{AC}^l$= -10 A, $I_{DC}^l = 35\,A$. Calculate the AC and DC components of the summative current and the current going to the AC grid.

[2]

iii) Explain briefly the role of the summative current in the operation of the converter.

[2]

iv) Write the differential equations that relate the grid and stack voltages (ie V_{AC}, V_{DC}, V_u, V_l) and the arm currents (I_u, I_l) in the circuit. Does V_u affect I_l?

[2]

v) Is it possible to make the converter draw more power from the positive pole of the DC grid (ie the upper $V_{DC}/2$ in the diagram) than from the negative pole (ie the lower $V_{DC}/2$)? Explain what current the controller would use. Is there any reason why the controller shouldn't do it considering the effect this would have on the AC grid?

[2]

vi) List the benefits and the drawbacks of using full-bridge submodules instead of half-bridge submodules.

[2]